Gulf of Alaska and Aleutian Islands Bottom Trawl Surveys
Gear Description for Poly Nor'Eastern Trawl

Netting: Polyethylene, 5" stretch measure - 4 mm top and sides, 5 mm bottom and intermediate.

Headrope: 89' 1" of ½" (6×19) galvanized wire rope wrapped with ⅜" polypropylene rope.
Both eyes have ½" gusseted thimble; 89' 1" doesn't include the length of either eye.
Headrope length is measure from the top of the nicro sleeve to the top of the nicro at other end.
Top wing is hung over 16.31" per taper combination.
Headrope setback - 18" of ½" long link galvanized alloy chain and connecting hardware.

Bolsh line: 81' 7" plus thimbled eyes of ⅜" (6×19) galvanized wire rope, wrapped with ⅜" polypropylene rope.

Fishing Line: 81' of ½" long link galvanized alloy chain.
Safe working load of 11,300 lbs.

Breastlines: ½" (6×19) galvanized wire rope wrapped with ⅜" polypropylene rope.
Top corner 19' 6", bottom corner 8' 8", bottom side panel 30' 6".
All lengths are measured from the top of nicro sleeve at wing tip and including thimbled eye at ribline.

Riblines: ¾" Samson 2in1® Duralon braided trawl rope.
Riblines are hung at 98% of stretch measure of gored seam.
All measurements are made with 400 lbs of tension on rope.
Gored seams are attached to the ribline using white untreated 60 T braided nylon hanging twine used to tie a benzel every 16".

Roller Gear: 79' 6" bearing to bearing (eye to eye). ¾" (6×19) galvanized wire rope with 14" bobbins (Footrope) and 4" discs.
See attached diagram for more detailed description.

Side Seams: Side seams are laced in each panel individually.
Seams are made by gathering 3 meshes (4 knots) and lacing them together using white, double 21 thread perma-grip® (or like kind) nylon, three-strand twine.
Individual panels are laced together using green, double 21 thread perma-grip® nylon, three-strand twine.
Panels which are secured to framing lines have selvage edge created by gathering 3 meshes as described above.
These selvage edges are hung tight to framing lines using 60 T braided nylon hanging twine.

Flotation: Twenty 12" Cycolac® side lug trawl floats (23 lbs of buoyancy each) hung evenly to headrope starting 1' from wing tip, and four 8" side lug trawl floats (5.4 lbs buoyancy each) hung two each on both sides of center, leaving 3' open in the center of the headrope.
Total buoyancy - 486 lbs.

Splitting Gear: ¾" Spectra® rope spliced 21' with eyes in each end.
The rope is passed through four galvanized rings secured to each ribline 18 meshes from the bottom. A second identical splitting strap is located 38 meshes from the bottom.

Codend: 3½" stretched measure (including 1 knot) polyethelene 4 mm Double Bar mesh.
Four panels cut 30 meshes long by 100 meshes deep.
Two meshes in a gored seam each side, leaving 26 open meshes per panel.
Gored seams are laced together using double 21 T nylon twine.
Riblines in codend are ¾" 2in1® Duralon braided trawl rope, hung at 90% of the stretch measure of the gored seams.
Codend is closed at terminal end using 25 2½" X ¼" galvanized steel rings.
A ¼" Duralon rope is passed through the selvage mesh and a ring is attached to the rope using a cow hitch every 12", with five open meshes between each ring.
The bag is then closed using a ⅝"-¾" hauling clip.

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Materials List for Poly Nor'Eastern Trawl

Specifications of wire, chain, and rope include rated breaking strength (BS), specific gravity (SG), and density.

WEB: All web in the trawl should be depth stretched and heat set in the manufacturing process. Mesh sizes listed in the plan are given in "stretched measure", a standard method of measuring mesh size that includes the length of one knot.
Top, body and wings, side panels ~ 5", 4-mm polyethylene knotted web - orange.
Bottom, body and wings, intermediate ~ 5", 5mm polyethylene knotted web - orange.
Liner ~ 1¼", # 18T three strand nylon knotted web - dyed green.
Chaffing gear ~ 6", 6mm double bar mesh, polyethylene knotted web - orange.
Chaffing strip, bottom~ 5", 5mm double bar mesh, polyethylene knotted web - orange.

WIRE ROPE: ½" 6×19 galvanized fiber core (BS 9,700 kg, 0.62 kg/m), eyes are formed using galvanized reinforced thimbles.

CHAIN: ½" (16 mm) galvanized, long link marine deck lashing chain, grade 70 (BS 20,500 kg, 3.42 kg/m).
⅝" (16 mm) galvanized, long link marine deck lashing chain, grade 70 (BS 28,670 kg, 5.06 kg/m).

FLOATS: Twenty 12" side lug trawl floats (23 lbs. buoyancy each) depth rated to a minimum of 450 fm. Four 8" side lug trawl floats (5.4 lbs buoyancy each).

RIBLINES: ¾" Samson Duralon™ 2-in-1® stable braid, white, light bonding (BS 8,800 kg, SG 1.83, 0.27 kg/m).

TWINE: Sewing panels together: 5-mm polyethylene twine, any color other than orange.
Lacing seams: # 21T white, three strand nylon.
Joining top and side seams together: # 21T green three strand nylon.
Joining bottom and intermediate seams: 60T white braided hanging twine.

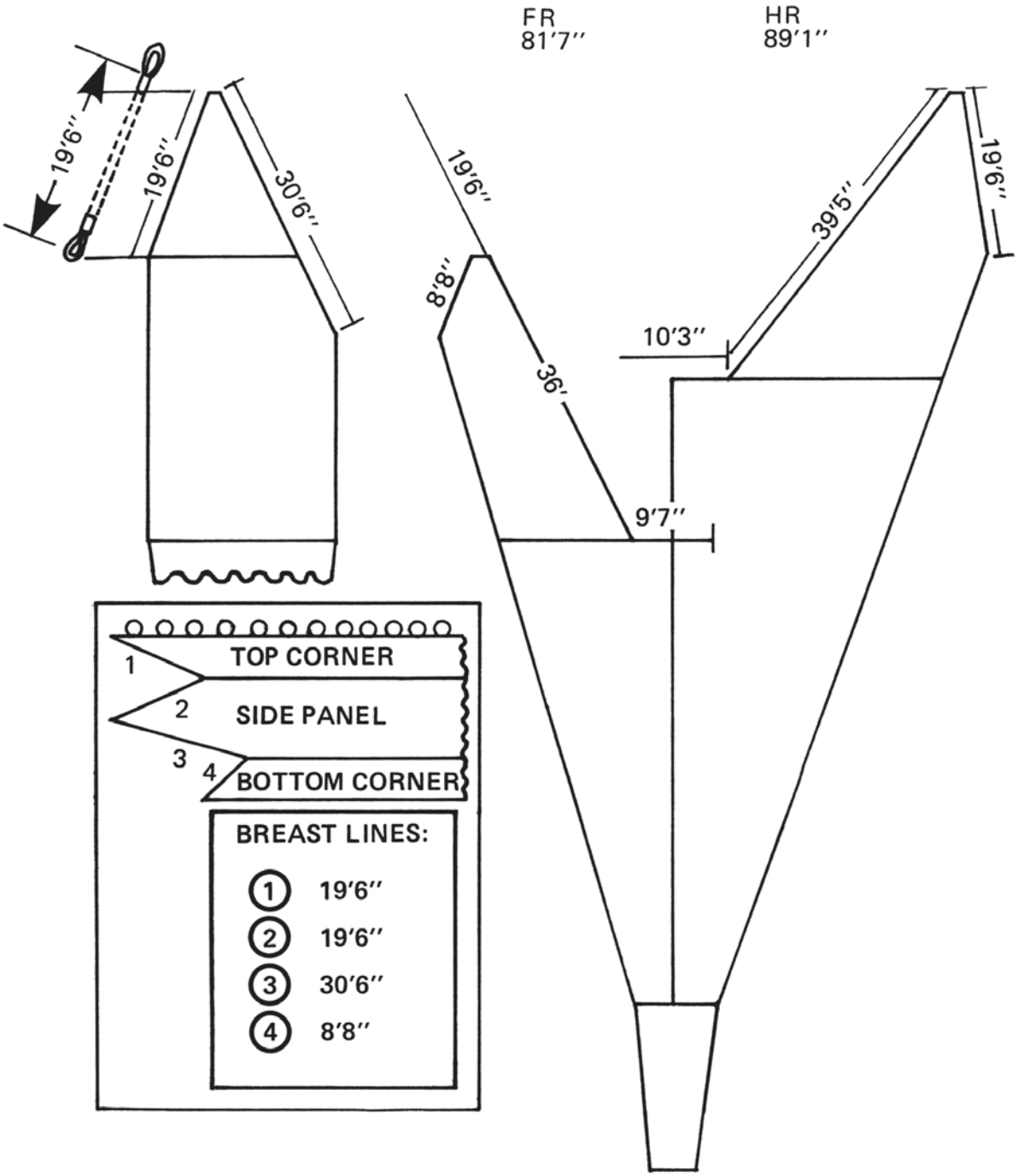
ROPE:

Float lines - 1/4" polypropylene (BS 570 kg, SG 0.95).
Pucker ring lines- Samson 1/4" Duralon™ 2-in-1®(BS 1,270 kg, SG 1.38)
Restrictor rope- 1 1/8" Poly Plus™ three strand (BS 10,740 kg, SG 0.91).
Splitting strap- 3/4" Spectra™ braided rope with eye each end (BS 21,773 kg, SG 0.98).
Rope to serve cables- 3/8" polypropylene three strand.
Chaffing gear- Poly hula skirt material.

FOOTROPE: (GOA, Aleutian Islands, MACE)

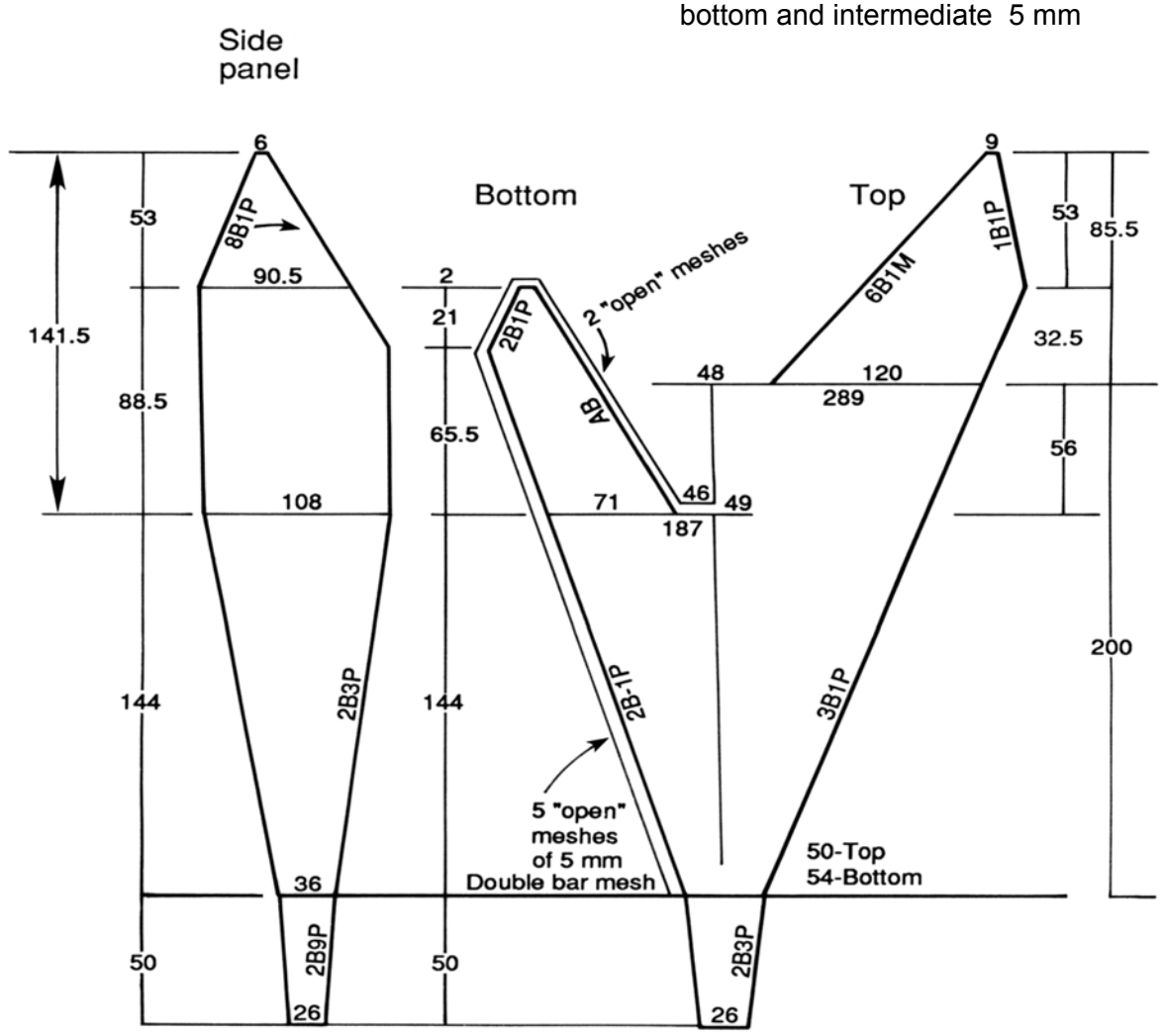
¾" 6×19 galvanized, fiber core wire rope 79' 6", (BS 21,600 kg, 1.41 kg/m) .
4" rubber disc with a 1" hole- wing extensions, center section between bobbins
8" rubber disc with a 2¼" hole - wing extensions.
⅝" Long link marine deck lashing chain, grade 70.
Laminated rubber "cone style" wing bobbins with a 2½" steel reinforced hole.
Chain droppers 10" in length. Five links of ⅜" chain and one ring
2½"×¼"
¾" barrel clamps
Steel plate wire rope washers- 5" ×⅜"- 1" center hole.
1¼" ID - 1¾" OD heavy hose over bare ¾" wire.
Four hole triangle delta plate, ⅜" steel.

Gulf of Alaska and Aleutian Islands Bottom Trawl Surveys
Framing Lines for Poly Nor'Eastern Trawl



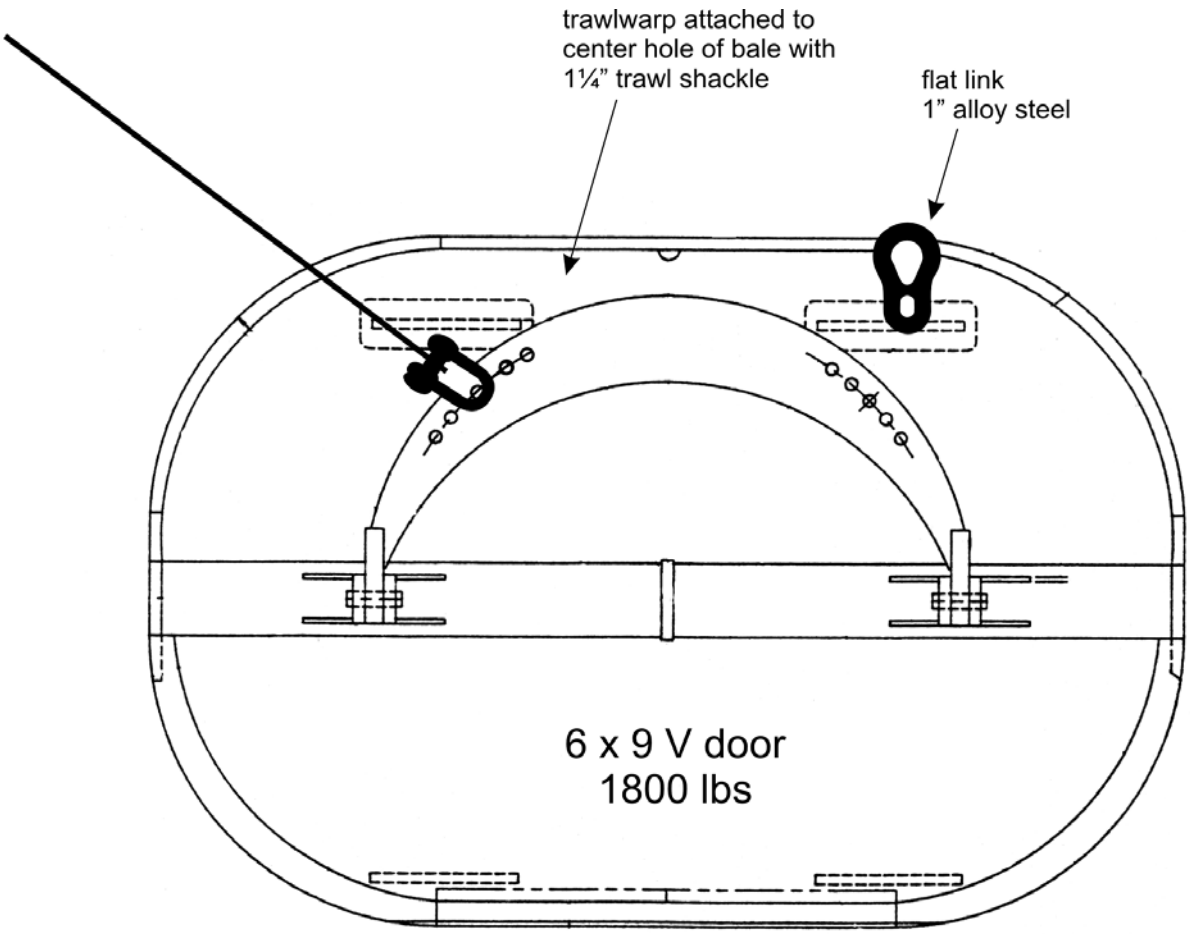
Gulf of Alaska and Aleutian Islands Bottom Trawl Surveys
Net Plan for Poly Nor'Eastern Trawl
("Cut Plan" = Total Mesh Counts)

Twine Sizes: top and sides 4 mm
bottom and intermediate 5 mm

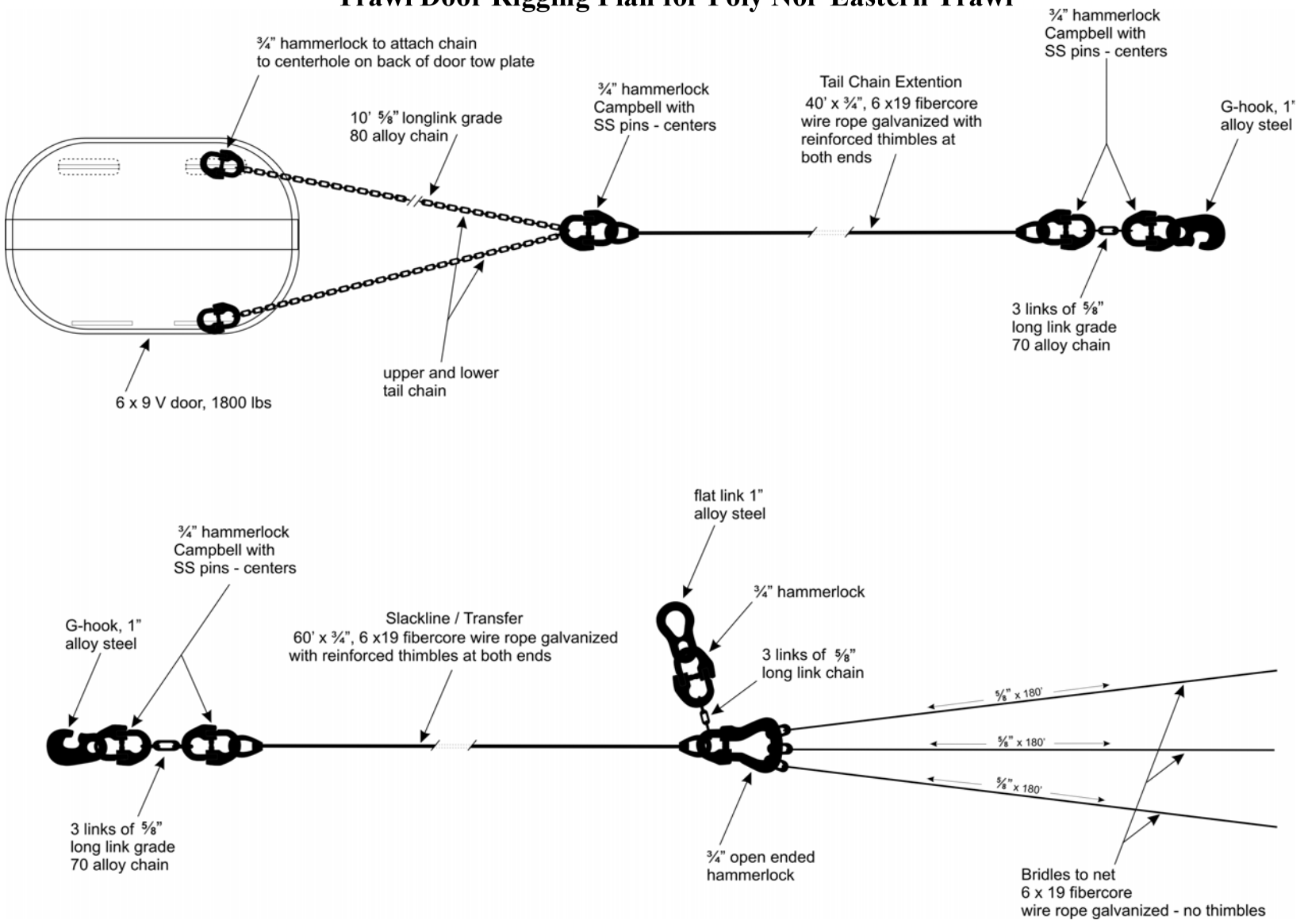


Web: Chaffing strip along inside of Bottom wings and Busom. Cut 8 meshes wide. 5 mm Double Bar mesh, goring 3 meshes on each side (leaving 2 open meshes). Secure 3 mesh of gore on inside (Bar Cut) of Bottom wings, and securing other gore to footrope (Bolsh).

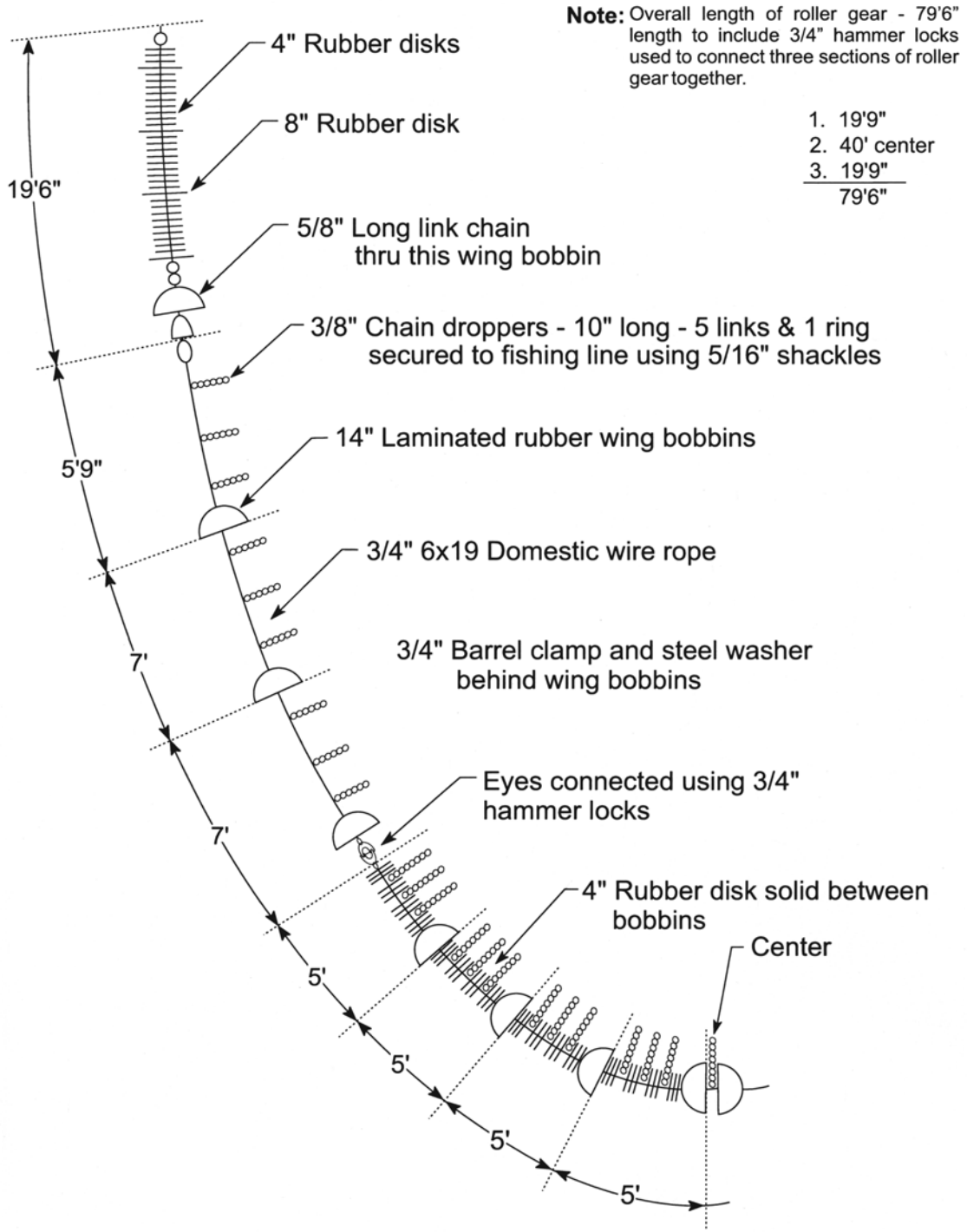
Gulf of Alaska and Aleutian Islands Bottom Trawl Surveys
Trawl Door Rigging Plan Detail for Poly Nor'Eastern Trawl
Sole Manufacturer NET Systems, Inc., Bainbridge Island, WA



Gulf of Alaska and Aleutian Islands Bottom Trawl Surveys
Trawl Door Rigging Plan for Poly Nor'Eastern Trawl



Gulf of Alaska and Aleutian Islands Bottom Trawl Surveys
Bobbin Roller Gear Construction Plan for Poly Nor'Eastern Trawl



Vessel	
Cruise	
Date	
Checked by	
Checked by	
Net I.D. number	

Survey Trawl Check List

Gulf of Alaska and Aleutian Islands Bottom Trawl Surveys

Poly Nor'Eastern Trawl

